

Chen Hajaj - January 2026

CONTACT INFORMATION Department of Industrial Engineering and Management Ariel University
<https://www.ariel.ac.il/wp/chen-hajaj/> chenha@ariel.ac.il

RESEARCH INTERESTS Machine Learning, Medical Data Science, Cybersecurity, Encrypted Traffic Classification, E-commerce, Behavioral Game Theory, Crowdsourcing

POSITIONS

Director, Data Science Research Center 2018 – 2024

Chair, Faculty of Engineering Review Board 2022 – present

Faculty 2018–present
Industrial Engineering and Management, Ariel University
School of Computer Science, Ariel University

Data Science Fellow, Vanderbilt University 2017 – 2018
Data think tank - Tennessee Educational Research Alliance (TERA)
Data think tank - Anthropogenic landscape features in South America

Postdoc Researcher 2016 – 2018
Electrical Engineering and Computer Science, Vanderbilt University
Host: Vorobeychik Yevgeniy, Ph.D

EDUCATION

Bar-Ilan University, Ramat Gan, Israel
Ph.D., Computer Science 2016
Thesis Topic: *Intelligent Mechanisms for Platforms in Modern Markets*
Advisors: Sarne David, Ph.D. and Hassidim Avinatan, Ph.D.
M.Sc., Electrical Engineering 2012
Topic: *Three-Dimensional Group Registration of Mesh Objects*
Advisor: Keller Yosi, Ph.D.
B.Sc., Computer Engineering 2010

REFEREED JOURNAL PUBLICATIONS

1. Ron Hirshprung, Ori Lesman, **Chen Hajaj**. The Sound of Emotions, An Artificial Intelligence Approach to Predicting Emotions from Musical Selections. In *Multimedia Systems*. 2026
2. Anat Goldstein, **Chen Hajaj**, Pierre Singer, Orit Raphaeli. Survival Forest Models for ICU Mortality Prediction based on Nutrition and Clinical Factors. In *IEEE Access*. 2026
3. Zafir Avrahami, Moti Zwilling, **Chen Hajaj**. Leveraging OSINT for Advanced Proactive Cybersecurity: Strategies and Solutions. *IEEE Access*. 2025.
4. Ro'i Lang, Or Haim Anidjar, Sahar Slonimsky, **Chen Hajaj**, Oz Golan, Carmel Matias, Alex Diskin, Evgeny Strokin, Mor Mega. Measuring and Analyzing Defects of Additive Manufactured Ti-6Al-4V Specimens Through Image Segmentation. *Fatigue & Fracture of Engineering Materials & Structures*. 2025.

5. Udi Aharon, Ran Dubin, Amit Dvir, **Chen Hajaj**. A classification-by-retrieval framework for few-shot anomaly detection to detect API injection. In *Computers & Security*. 2025
6. Or Haim Anidjar, Revital Marbel, Ran Dubin, Amit Dvir, **Chen Hajaj**. Extending Limited Datasets with GAN-Like Self-Supervision for SMS Spam Detection. In *Computers & Security*. 2024
7. Assaf Frajman, Shimon Izhakian, Ori Mekiten, Ori Hadar, Ariel Lichtenstadt, **Chen Hajaj**, Shon Schori, Moshe Heching, Dror Rosengarten and Mordechai R. Kramer. Phenotypical Characteristics of Nontuberculous Mycobacterial Infection in Patients with Bronchiectasis. In *Respiratory Research*. 2024
8. **Chen Hajaj**, Porat Aharon, Ran Dubin, and Amit Dvir. The Art of Time-Bending: Data Augmentation and Early Prediction for Efficient Traffic Classification. In *Expert Systems with Applications*. 2024
9. Anat Goldstein and **Chen Hajaj**. Measuring flight-destination similarity: A multidimensional approach. In *Expert Systems with Applications*. 2024
10. Ofek Bader, Adi Lichy, Amit Dvir, Ran Dubin, and **Chen Hajaj**. OSF-EIMTC: An Open-Source Framework for Standardized Encrypted Internet Traffic Classification. In *Computers Communications*. 2024
11. Orit Raphaeli, Liran Statlender, **Chen Hajaj**, Itai Bendavid, Anat Goldstein, Eyal Robinson, and Pierre Singer. Using Machine-Learning to Assess the Prognostic Value of Early Enteral Feeding Intolerance in Critically Ill Patients: A Retrospective Study. In *Nutrients*. 2023
12. Harel Berger, Amit Dvir, Enrico Mariconti, and **Chen Hajaj**. Breaking the structure of MaMaDroid. In *Expert Systems with Applications*. 2023
13. Ofir Joseph, Avshalom Elmalech, and **Chen hajaj**. Detecting Parallel Covert Data Transmission Channels in Video. In *Electronics*. 2023
14. Or Haim Anidjar, Yannick Estève, **Chen Hajaj**, Amit Dvir, Itshak Lapidot. Speech and Multilingual Natural Language Framework for Speaker Change Detection and Diarization. In *Expert Systems With Applications*. 2023
15. Adi Lichy, Ofek Bader, Ran Dubin, Amit Dvir, **Chen Hajaj**. When a RF Beats a CNN and GRU, Together - A Comparison of Deep Learning and Classical Machine Learning Approaches for Encrypted Malware Traffic Classification. In *Computers & Security*. 2023
16. Uzi Ben-Shalom, **Chen Hajaj**, Nitza Davidovich, Corine Berger. The Online temporary learning groups in higher education: Interactions, compensation, and maximization of achievements in an Israeli case study. In *Journal of Education Culture and Society*. 2023
17. Rotem Bar and **Chen Hajaj**. SimCSE for Encrypted Traffic Detection and Zero-Day Attack Detection. In *IEEE ACCESS*. 2022

18. **Chen Hajaj**, Nitay Hason, and Amit Dvir. Less is More: Robust and Novel Features for Malicious Domain Detection. In *Electronics*. 2022
19. Anat Goldstein and **Chen Hajaj**. The Hidden Conversion Funnel of Mobile vs. Desktop Consumers. In *Electronic Commerce Research and Applications (ECRA)*, 2022
20. Corine Berger, Uzi Ben-Shalom, **Chen Hajaj**, Nitza Davidovich. Patterns of Activity in Temporary Digital Learning Groups, and Its Correlation With Academic Achievement Among University Students in an Israeli University. In *Social Issues in Israel*. 2022
21. Harel Berger, **Chen Hajaj**, and Amit Dvir. Crystal Ball: From Innovative Attacks to Attack Effectiveness Classifier. *IEEE ACCESS*. 2021
22. Ron Hirschprung and **Chen Hajaj**. Prediction Model for the Spread of the COVID-19 Outbreak in the Global Environment. *Heliyon*. 2021
23. Or Anidjar, Itshak Lapidot, **Chen Hajaj**, Amit Dvir and Issachar Gilad. Hybrid Speech and Text Analysis Methods for Speaker Change Detection. In *IEEE/ACM Transactions on Audio Speech and Language Processing*. 2021
24. **Chen Hajaj**, Zlatko Joveski, Sixie Yu, Yevgeniy Vorobeychik. Robust Coordination in Adversarial Social Networks: From Human Behavior to Agent-Based Modeling. in *Network Science*, 2021.
25. Amit Dvir, Angelos K. Marnerides, Ran Dubin, Nehor Golan, and **Chen Hajaj**. Encrypted Video Traffic Clustering Demystified. In *Computers & Security*, 96 ,2020.
26. **Chen Hajaj** and David Sarne. Selective Opportunity Disclosure at the Service of Strategic Information Platforms. In *Autonomous Agents and Multi-Agent Systems (JAAMAS)*, 31(5):1133–1164, 2017.
27. **Chen Hajaj**, Noam Hazon, and David Sarne. Enhancing Comparison Shopping Agents through Ordering and Gradual Information Disclosure. In *Autonomous Agents and Multi-Agent Systems (JAAMAS)*, 31(3):696–714, 2017.
28. **Chen Hajaj**, Noam Hazon, and David Sarne. Improving Comparison Shopping Agents’ Competence through Selective Price Disclosure. In *Electronic Commerce Research, and Applications (ECRA)*, 14(6) 563–581, 2015.

REFEREED
CONFERENCE
PUBLICATIONS

1. Meir Litman and **Chen Hajaj**.Cleaner Adversarial CAPTCHAs: Intelligent Targets and Precise Noise for Usable Security. In *The 25th International Conference on Autonomous Agents and Multiagent Systems (AAMAS)* .2026
2. Revital Marbel, Yanir Cohen, Ran Dubin, Amit Dvir and **Chen Hajaj**. Cloudy with a Chance of Anomalies: Dynamic Graph Neural Network for Early Detection of Cloud Services’ User Anomalies. In *the 34th International Conference on Computer Communications and Networks (ICCCN)*. 2025

3. Simona Lisker, Ayelet Botman, **Chen Hajaj**, Ran Dubin, Amit Dvir. Optimized File Type Detection and One-Shot Retrieval. In *the IEEE International Conference on Communications (ICC)*. 2025
4. Zohar Simhon, Matan Weiss, **Chen Hajaj**, Revital Marbel, Ran Dubin, Amit Dvir. A New D-MAGIC: Dynamic Model for Cybersecurity Attack Detection Using GNNs into Clustering. In *Proceedings of the IEEE International Conference on Communications (ICC)*. 2025
5. Yehonatan Zion, Porat Aharon, Ran Dubin, Amit Dvir, **Chen Hajaj**. Enhancing Encrypted Internet Traffic Classification Through Advanced Data Augmentation Techniques. In *Proceedings of the IEEE International Conference on Communications (ICC)*. 2025
6. Angelos Marnerides, **Chen Hajaj**, Revital Marbel, Ran Dubin, Amit Dvir. PQClass: Classification of Post-Quantum Encryption Applications in Internet Traffic. In *Proceedings of the IEEE International Conference on Communications (ICC)*. 2025.
7. Anat Goldstein and **Chen Hajaj**. Warm Recommendation: Enhancing Cold Start Recommendations Using Multimodal Product Representations. In *International conference of the Association for Information Systems (ICIS)*. 2024
8. Nathan Dillbary, Roi Yozevitch, Amit Dvir, Ran Dubin, and **Chen Hajaj**. In *Hidden in Time, Revealed in Frequency: Spectral Features and Multiresolution Analysis for Encrypted Internet Traffic Classification*. In *the Proceedings of IEEE Consumer Communications & Networking Conference*. 2024
9. Anat Goldstein and **Chen Hajaj**. Warming Up the Cold Start: A Multimodal Approach. In *Special Interest Group on Decision Support and Analytics (SIGDSA) Symposium*. 2023
10. Orit Raphaeli, Liran Statlander, **Chen Hajaj**, Itai Ben David, and Pierre Singer. Protein Intake and Clinical Outcomes of Critically Ill Patients. In *the 44th European Congress for Clinical Nutrition and Metabolism (ESPEN)*. 2022
11. Anat Goldstein and **Chen Hajaj**. Measuring Product Similarity: A Multidimensional Approach. In *the Eighteenth Symposium on Statistical Challenges in Electronic Commerce Research*. 2022
12. Ofek Bader, Adi Lichy, **Chen Hajaj**, Ran Dubin, and Amit Dvir. MalDIST: From Encrypted Traffic Classification to Malware Traffic Detection and Classification. In *the Proceedings of IEEE Consumer Communications & Networking Conference*. 2022
13. Orit Raphaeli, **Chen Hajaj**, Itai Ben David, Anat Goldstein, Elad Chen, Pierre Singer, and Uri Gabbay. Using the Cardio-Vascular Index (CVRI) To Predict Mortality in Septic Shock. In *the 40th International Symposium on Intensive Care & Emergency Medicine*. 2022
14. Orit Raphaeli, **Chen Hajaj**, Itai Ben David, Anat Goldstein, Elad Chen, Pierre Singer. Using Machine Learning to Support Early Prediction

- of Feeding Intolerance in Critically Ill Patients. In *Intensive Care Medicine Experimental*. 2021
15. Orit Raphaeli, **Chen Hajaj**, Itai Ben David, Anat Goldstein, Elad Chen, Pierre Singer. Using Machine Learning to Compare Gastric Residual Volume Thresholds as Predictors of Clinical Outcomes in Critically Ill Patients. In *International Symposium of Intensive Care and Emergency Medicine*. 2021
 16. Or Anidjar, Itshak Lapidot, **Chen Hajaj** and Amit Dvir. A Thousand Words are Worth More Than One Recording: Word-Embedding Based Speaker Change Detection. In *Interspeech*. 2021
 17. Orit Raphaeli, **Chen Hajaj**, Itai Ben David, Anat Goldstein, Elad Chen, Pierre Singer. Feeding Intolerance as a Predictor of Clinical Outcomes in Critically Ill Patients: A Machine Learning Approach. In *the 43rd European Society for Clinical Nutrition and Metabolism (ESPEN)*. 2021
 18. Uzi Ben Shalom, **Chen Hajaj**, Nitza Davidovich and Corinne Berger. Online temporary learning groups in higher education – Interactions, compensation, and maximization of achievements in an Israeli case-study. In *the 1st international e-conference on teacher education generation next: perspectives, opportunities & challenges*. 2021
 19. Vitali Demianiuk, **Chen Hajaj**, and Kiril Kogan. PCL: Packet Classification with Limited Knowledge. *INFOCOM*. 2021
 20. Anat Goldstein and **Chen Hajaj**. The Different Path to Purchase of Mobile and Desktop Consumers: Analyzing Consumers' Progress in the Conversion Funnel Using Hidden Markov Models. In *Statistical Conference in E-Commerce Research (SCECR)*. 2020
 21. Nitay Hason, Amit Dvir, and **Chen Hajaj**. Robust Malicious URL Detection. In *the 4th International Symposium on Cyber Security Cryptology and Machine Learning (CSCML)*. 2020
 22. Liang Tong, Bo Li, **Chen Hajaj**, Chaowei Xiao, Ning Zhang, Yevgeniy Vorobeychik. Improving Robustness of ML Classifiers against Realizable, Evasion Attacks Using Conserved Feature. In *the 28th USENIX Security Symposium*. 2019
 23. Yevgeniy Vorobeychik, Myrna Wooders, Greg Leo, Matthew Chambers, **Chen Hajaj**, Jian Lou, and Martin Van der Linden. Non-Cooperative Team Formation and a Team Formation Mechanism. In *the 11th annual conference of the Israeli chapter of the game theory society*. 2019
 24. **Chen Hajaj**, Sixie Yu, Zlatko Joveski, Yifan Guo, and Yevgeniy Vorobeychik. Adversarial Coordination on Social Networks. In *the International Conference on Autonomous Agents and Multiagent Systems (AAMAS)*. 2019
 25. Cheng Ye, Joseph Coco, Anna Epishova, **Chen Hajaj**, Henry Bogardus, Laurie Novak, Joshua Denny, Yevgeniy Vorobeychik, Thomas Lasko, Bradley Malin, and Daniel Fabbri. Crowdsourcing Clinical Chart Reviews. In *AMIA Joint Summits on Translational Science*. 2018

26. **Chen Hajaj** and Yevgeniy Vorobeychik. Adversarial task assignment. In *International Joint Conference on Artificial Intelligence and European Conference on Artificial Intelligence (IJCAI-ECAI)*. 2018
27. Cheng Ye, Joseph Coco, Anna Epishova, **Chen Hajaj**, Henry Bogardus, Laurie Novak, Joshua Denny, Yevgeniy Vorobeychik, Thomas Lasko, Bradley Malin and Daniel Fabbri. A Crowdsourcing Framework for Medical Data Sets. In *American Medical Informatics Association (AMIA)*. 2018
28. Avshalom Elmalech, David Sarne, Esther David, and **Chen Hajaj**. Enhancing Crowdworkers' Vigilance. In *International Joint Conference on Artificial Intelligence (IJCAI)*. 2017
29. Avshalom Elmalech, David Sarne, Esther David, and **Chen Hajaj**. Extending Workers' Attention Span Through Dummy Sub-Goals. In *AAAI Conference on Human Computation and Crowdsourcing (HCOMP)*. 2016. **Best paper finalist**.
30. **Chen Hajaj**, John P. Dickerson, Avinatan Hassidim, Tuomas Sandholm and David Sarne. Strategy-Proof and Efficient Kidney Exchange Using a Credit Mechanism. In *AAAI Conference on Artificial Intelligence (AAAI)*. 2015
31. Tal Darom, Yaniv Gur, **Chen Hajaj** and Yosi Keller. Automatic Anatomical Shape Correspondence and Alignment Using Mesh Features. In *International Symposium on Biomedical Imaging (ISBI)*. 2015
32. **Chen Hajaj** and David Sarne. Strategic Information Platforms - Selective Disclosure and The Price of "Free". In *ACM Conference on Economics and Computation (EC)*. 2014
33. **Chen Hajaj**, Noam Hazon, and David Sarne. Ordering Effects and Belief Adjustment in the Use of Comparison Shopping Agents. In *AAAI Conference on Artificial Intelligence (AAAI)*. 2014
34. **Chen Hajaj**, Noam Hazon, David Sarne, and Avshalom Elmalech. Search More, Disclose Less. In *AAAI Conference on Artificial Intelligence (AAAI)*. 2013

SHORT PAPERS
AND REFEREED
WORKSHOPS

1. Brigitta R Schvarcz and **Chen Hajaj**. Exploring Student Perspectives of the Impact of VR in ESP. In *TESOL*. 2024
2. Bar Vinder, **Chen Hajaj**, and Idit Avrahami. A method for evaluating the performance of a novel percutaneous Left Ventricular Assist Device and optimizing its design. In *VPH 2022 PHYSIOLOGICAL HUMAN CONFERENCE*. 2022
3. Harel Berger, **Chen Hajaj**, and Amit Dvir. Evasion is not enough: A case study of Android malware. In the *4th International Symposium on Cyber Security Cryptology and Machine Learning (CSCML)*. 2020
4. **Chen Hajaj** and Yevgeniy Vorobeychik. Adversarial Randomized Task Assignment, 6th Day on Computational Game Theory. 2019

5. **Chen Hajaj**, Sixie Yu, Zlatko Joveski and Yevgeniy Vorobeychik. Adversarial Coordination on Social Networks. In *The AAMAS-IJCAI workshop on agents and incentives in artificial intelligence(AI³)*. 2018
6. Jian Lou, **Chen Hajaj**, Myrna Wooders and Yevgeniy Vorobeychik. Automated Mechanism Design for the Roommates Problem. In *The AAMAS-IJCAI workshop on agents and incentives in artificial intelligence (AI³)*. 2018
7. Jian Lou, Martin Van der Linden, Pranav Batra, **Chen Hajaj**, Greg Leo, Yevgeniy Vorobeychik and Myrna Wooders. Rotating Proposer Mechanisms for Team Formation. In *Workshop on Cooperative Games in Multiagent Systems (CoopMAS)*.2017. **Most visionary paper.**
8. **Chen Hajaj**, David Sarne, and Lea Perets. Advanced Service Schemes for a Self-Interested Information Platform. In *International Conference on Autonomous Agents and Multiagent Systems (AAMAS)*. 2014

BOOK CHAPTERS Sofia Ceppi, Esther David, **Chen Hajaj**, Valentin Robu, and Ioannis A. Vetsikas, editors, Agent-Mediated Electronic Commerce: Designing Trading Strategies and Mechanisms for Electronic Markets, Springer, LNBIP 271, 2017.

PATENTS **Chen Hajaj**, David Sarne, Noam Hazon. “Selective Options Disclosure Method for Affecting a User’s Decision-Making” **US Provisional Patent Application No. 61/842,974**
 Amir Glik, **Chen Hajaj**, Orit Rephaeli, Anat Goldstein. ”PREDICTING ONSET OF NEURODEGENERATIVE DISEASES” **US Provisional Patent Application No. 63/456,555**

STUDENTS

Ph.D. students

- | | |
|--------------------------------------|--------------|
| 1. Natan Dilbberry, Ariel University | 2024–present |
| 2. Ronyt Gomez, Bar-Ilan University | 2024–present |
| 3. Aviad Levi, Ariel University | 2023–present |
| 4. Eyal Paz, Ariel University | 2022–present |
| 5. Udi Aharon, Ariel University | 2022–present |
| 6. Yehonatan Zion, Ariel University | 2022–present |
| 7. Harel Berger, Ariel University | 2019–2022 |
| 8. Or Anidjar, Ariel University | 2019–2022 |

Pre-Ph.D.

- | | |
|-------------------------------------|-----------|
| 1. Zafir Avrahami, Ariel University | 2023-2024 |
| 2. Omri Arbiv, Ariel University | 2022-2024 |
| 3. Meir Litman, Ariel University | 2020-2023 |

Masters (Thesis)

1. Polina Korogodsky, Ariel University	2023–present
2. Porat Aharon, Ariel University	2022–2023
3. Ofir Yosef, Bar-Ilan University	2021–2023
4. Bar Vinder, Ariel University	2021–2023
5. Adi Lichi, Ariel University	2021–2023
6. Ofek Bader, Ariel University	2021–2022
7. Nitay Hason, Ariel University	2019–2020
8. Nabeeha Yassin, Ben-Gurion University	2018–2021

Masters (Final project)

1. Elad Chen, Ariel University	2021-2023
2. Yuval Ben Yaakov, Ariel University	2021-2023
3. Tal Abed, Ariel University	2021-2022
4. Ariel Yehezkel, Ariel University	2021-2022
5. Bar Ben David, Afeka College	2022-2024
6. Ben Ron Udi, Afeka College	2022-present
7. Chen Shiener, Afeka College	2022-2023
8. Nathanael Guinzburg, Afeka College	2022-2023
9. Michael Gudovsky, Afeka College	2022-2023
10. Rotem Bar, Afeka College	2021-2022
11. Pavel Shochet, Afeka College	2021-2022
12. Eliran Barai, Afeka College	2020–2021

Research assistants

1. Barak Wirzeberger, Ariel University	2024–present
2. Roni Epstein, Ariel University	2024–present
3. Shon Schori, Ariel University	2022–present
4. Elad Chen, Ariel University	2020–2021
5. Alon Trugman, Ariel University	2020–2021
6. Idan Masuri, Ariel University	2019–2020

Program Committee

ILAIS (2022,2023), ICIS (2022), AAAI (2013, 2014, 2017, 2018, 2019, 2023, 2024), AAMAS (2013, 2014, 2015, 2017, 2018, 2019, 2020), ACM-EC (2018), AMIA (2019, 2020, 2021, 2022), AMEC (2013, 2015), CPSCom (2017), GameSec (2016), ICAART (2020), IJCAI (2016, 2017, 2018, 2019, 2022), SODA (2015), ECAI (2024).

Reviewer

Annals of Mathematics and Artificial Intelligence, Journal of Information Security and Applications (JISA), Artificial Intelligence Review (AIRE), Journal of Artificial Intelligence Research (JAIR), Computers & Security, Expert Systems with Applications, Computer Networks, Health Systems, Transactions on Knowledge and Data Engineering.